



Tribhuvan University

Faculty of Humanities and Social Sciences

KHOJA (Smart Tourism and Bus Navigation System)

A PROJECT PROPOSAL

**Submitted to
Department of Computer Application
DAV College**

In partial fulfillment of the requirements for the Bachelors in Computer Application

Submitted by
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Tribhuvan University
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SUPERVISOR'S RECOMMENDATION

I hereby recommend that this project prepared under my supervision by AVASH KHADKA entitled “**KHOJA - SMART TOURISM AND BUS NAVIGATION SYSTEM**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Applications is recommended for the final evaluation.

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LETTER OF APPROVAL

This is to certify that this project prepared by AVASH KHADKA entitled “**KHOJA - SMART TOURISM AND BUS NAVIGATION SYSTEM**” in partial fulfillment of the requirements for the degree of Bachelor of Computer Applications has been evaluated. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

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ABSTRACT

The **KHOJA** is a web-based application designed to enhance the travel experience by integrating tourism information, location-based navigation, and real-time public transportation services into a single platform. The system enables users to discover nearby tourist attractions, view detailed information such as descriptions, images, and categories, and navigate efficiently using GPS and interactive maps. It also provides real-time bus tracking, route information, and estimated arrival updates to improve the accessibility and convenience of public transportation. A community contribution feature allows users to submit new tourist locations, reviews, and reports, while an administrative verification process ensures the accuracy and reliability of the published information. Developed using HTML, CSS, JavaScript, PHP, MySQL, Leaflet, OpenStreetMap, and WebSocket technology, the system aims to promote smart and sustainable tourism by simplifying travel planning, improving navigation, encouraging community participation, and providing an efficient, reliable, and user-friendly solution for both tourists and local travelers.

Keywords: *Smart Tourism, Navigation System, Real-Time Bus Tracking, Location-Based Services (LBS), Geographic Information System (GIS), GPS Navigation, Community Contribution, Public Transportation, Interactive Maps, Tourism Management.*

ACKNOWLEDGEMENT

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LIST OF ABBREVIATION

API	Application Programming Interface
CSS	Cascading Style Sheets
GIS	Geographic Information System
GPS	Global Positioning System
HTML	HyperText Markup Language
HTTP	HyperText Transfer Protocol
JSON	JavaScript Object Notation
LBS	Location-Based Services
OSM	OpenStreetMap
PHP	Hypertext Preprocessor
SQL	Structured Query Language
UI	User Interface

CHAPTER 1: INTRODUCTION

1.1 Introduction

The rapid growth of urban transportation systems and digital mapping technologies has created opportunities for smarter and more connected travel experiences. People frequently face difficulties in finding reliable bus routes, tracking buses in real time, discovering useful nearby locations, and sharing local route information with others. Existing systems often focus only on tourism, ride-hailing, or static transport schedules, leaving a gap for a community-driven navigation and bus management platform.

The proposed project, **KHOJA: Smart Tourism Navigation and Bus Navigation System**, aims to address these challenges by combining real-time bus tracking, interactive route navigation, community location sharing, and role-based management portals for users, drivers, and administrators.

The system is designed as a modern web platform where, users can explore nearby places, view live buses on a map, and book transportation, drivers can register buses, create routes visually on a map, and manage trips, Administrators can moderate community contributions, manage buses and drivers, and monitor system analytics.

By integrating geolocation, mapping services, community contributions, and transport management into a single platform, **KHOJA** provides a smarter and more collaborative navigation ecosystem for everyday users, not just tourists.

1.2 Problem Statement

The current transportation and navigation ecosystem faces several challenges that reduce efficiency and convenience for users. One of the major issues is the lack of real-time visibility of public transport, as commuters often have no accurate information about the current location of buses or their expected arrival times. Additionally, navigation services, bus tracking systems, route-sharing platforms, and booking applications are typically developed as separate solutions, resulting in a fragmented user experience. Existing systems also provide limited opportunities for community participation, preventing users from contributing valuable local information such as newly established bus stops, road conditions, or important community locations.

Furthermore, drivers and transport operators often lack intuitive tools for creating, managing, and updating routes visually. Many local transportation platforms also suffer from outdated interfaces, incomplete information, and poor mobile responsiveness, leading to an unsatisfactory user experience. These limitations contribute to commuter inconvenience, inefficient transport coordination, and the underutilization of community-generated geographic information, highlighting the need for an integrated and user-friendly transportation and tourism mapping solution.

1.3 Objectives

1. To develop a web-based bus management system for managing bus routes, driver information, bus details, and transportation services.
2. To implement an interactive map-based navigation system that allows users to locate nearby places, view available bus routes, and track buses in real time.
3. To provide a platform for users to contribute and access location and transportation information, while enabling administrators and drivers to manage submitted locations, routes, and bus-related information.

1.4 Scope and Limitation

i. Scope

- Real-Time Bus Navigation – Users can track buses in real time, view available routes, and receive navigation assistance for traveling to tourist destinations using public transportation.
- Tourist Information Management – The system provides comprehensive information about tourist destinations, including descriptions, images, categories, operating hours, and other relevant details to help users plan their visits.
- Location-Based Services – The system utilizes GPS and interactive maps to display nearby attractions, calculate distances, and provide route guidance based on the user's current location.
- Community Contribution – Registered users can contribute new tourist locations, submit reviews and ratings, and report incorrect information, helping keep the platform updated and reliable.
- Administrative Management – Administrators can verify user contributions, manage tourist locations, users, bus routes, and system data through a centralized dashboard to ensure data accuracy and smooth operation.

ii. Limitation

- **Internet Dependency** – The system requires a stable internet connection to access real-time bus tracking, navigation, and updated tourism information.
- **GPS Accuracy** – The accuracy of navigation and nearby location recommendations depends on the availability and precision of GPS signals, which may vary in certain environments.
- **Manual Verification Process** – User-submitted locations and reports require administrative approval before becoming publicly available, which may introduce delays.
- **Limited Multi-Bus Route Calculation** – The current technology stack can efficiently calculate routes involving up to two bus transfers. Although supporting more than two bus switches is technically possible, the implementation is significantly more complex, and the current technology stack is not capable of handling such calculations efficiently.
- **Platform Availability** – The project is currently implemented as a web-based application and does not include a dedicated mobile application, which may limit accessibility for some users.

1.5 Methodology

Development of the **KHOJA** will follow the Waterfall Model of the Software Development Life Cycle (SDLC). The Waterfall Model is a linear and sequential model of software development. In this model, each phase of the SDLC which includes: Planning, Analysis, Design, Development and Implementation, must be completed before moving onto the next phase and returning to a previous phase is not recommended.

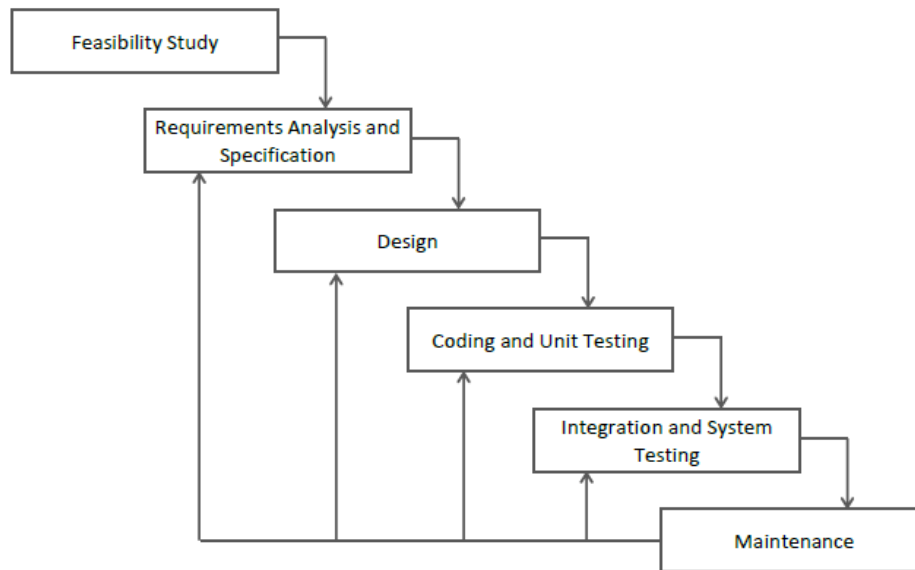


Figure 1. 1: Waterfall Model for KHOJA

This methodology is suitable for developing the **KHOJA** as the requirements, features, and roles in this system are fixed and clearly defined from the beginning. The Waterfall Model also provides a systematic and structured approach to software development, which allows for proper planning, clear documentation, and better scheduling of the entire project.

Another important advantage of this model is that it supports better project planning and estimation. Because the requirements of the system are already known, it becomes easier to define timelines, allocate resources, and manage tasks efficiently. This makes the development process more organized and predictable.

1.6 Report Organization

- **Chapter 1:**

This Chapter 1 includes the brief introduction, problem statement, objectives, methodology, scope and limitation of the project.

- **Chapter 2:**

The Chapter 2 involves the background study, literature review for the current existing system and Research regarding it.

- **Chapter 3:**

This includes system analysis and design of the Smart Tourism and Bus Navigation System which includes all the functional and non-functional requirements, feasibility analysis and Algorithm description.

- **Chapter 4:**

Implementation and Testing of the system such as unit testing.

- **Chapter 5:**

Completed Tasks and Remaining Tasks of the Smart Tourism and Bus Navigation System.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

Tourism is an important sector that contributes to economic development, employment opportunities, and cultural exchange. With the advancement of information and communication technology, digital platforms have become increasingly useful for tourists in discovering destinations, planning journeys, and accessing geographical information. Smart tourism combines digital technologies with tourism services to improve visitor experiences and destination management [2]. Interactive mapping technologies such as Leaflet provide web-based maps that allow users to view locations, markers, routes, and other geographical information [1]. OpenStreetMap also provides open and collaborative geographical data that can be used for mapping and location-based applications [3].

The proposed Smart Tourism and Navigation System integrates tourism information, interactive maps, location-based services, and public transportation into a single web-based platform. The system allows users to discover nearby tourist destinations, view their locations on an interactive map, calculate distances, and obtain navigation assistance. The system also focuses on bus navigation by providing bus-related information and supporting real-time bus location tracking. The Haversine formula is used to calculate the geographical distance between the user's current location and nearby tourist destinations [5]. In addition, Dijkstra's algorithm can be used to determine suitable shortest paths within a transportation network [4]. By combining these technologies, the proposed system aims to provide a convenient and efficient platform for tourism discovery and navigation.

2.2 Literature Review

Gretzel et al. introduced the concept of smart tourism as the use of information and communication technologies to support tourism experiences and destination management. Their study highlights the importance of digital technologies in providing tourists with information and improving the overall tourism experience [2]. This concept provides a foundation for developing systems that integrate tourism information with location-based and navigation services.

Leaflet is an open-source JavaScript library designed for interactive web mapping. It supports features such as map markers, zooming, panning, popups, and geographical

visualization, making it suitable for developing location-based web applications [1]. Similarly, OpenStreetMap provides freely available geographical data created and maintained through collaborative mapping. It can be integrated with mapping libraries such as Leaflet to display roads, locations, and other geographical features [3]. These technologies form the mapping foundation of the proposed Smart Tourism and Navigation System.

For route navigation, Dijkstra's algorithm is a well-known shortest-path algorithm used to determine the shortest path between nodes in a weighted graph. Dijkstra introduced the algorithm for solving shortest-path problems in graph networks [4]. In the proposed system, shortest-path concepts can be applied to transportation and navigation to help determine suitable routes between locations. However, complex routes involving multiple bus transfers can increase the difficulty of route calculation and management.

The Haversine formula is commonly used to calculate the great-circle distance between two points on the Earth's surface using their latitude and longitude coordinates [5]. In the proposed system, this formula is used by the Discover module to determine the distance between the user's current location and tourist destinations. The calculated distances allow the system to identify and display nearby locations based on the user's geographical position.

Tripadvisor is a tourism-focused platform that provides information about destinations, attractions, restaurants, hotels, reviews, ratings, and travel experiences [6]. User-generated reviews and ratings provide valuable information for travelers when selecting destinations and services. However, platforms primarily focused on tourism information may not provide specialized local bus tracking, bus route management, or community-based transportation services. The proposed system attempts to address this gap by combining tourist location discovery, community contribution, interactive navigation, and bus-related services within a single platform.

CHAPTER 3: SYSTEM ANALYSIS AND DESIGN

3.1 System Analysis

3.1.1 Requirement Analysis

i. Functional Requirements

S.N.	Requirement	Description
1	User Registration and Authentication	The system allows users to create accounts and securely log in using their personal information.
2	Location Detection using GPS	The system uses GPS to identify the user's current location accurately. It helps provide location-based services such as nearby places and available routes.
3	Nearby Place Discovery	The system allows users to find nearby tourist attractions based on their location. It provides details such as descriptions, images, opening hours, and entry fees of places.
4	Real-Time Bus Tracking	The system tracks buses using GPS data updated by drivers. Users can view current bus locations, routes, and estimated arrival information.
5	Community Location Contribution and Reporting	Users can submit requests for adding new tourist places or reporting incorrect information. Administrators review these requests before updating the system database.
6	Driver Bus Registration and Route Creation	Drivers can register buses and provide necessary bus information in the system. They can create routes and update details such as distance, destination, and travel time.

Table 3. 1: Functional Requirements of the system

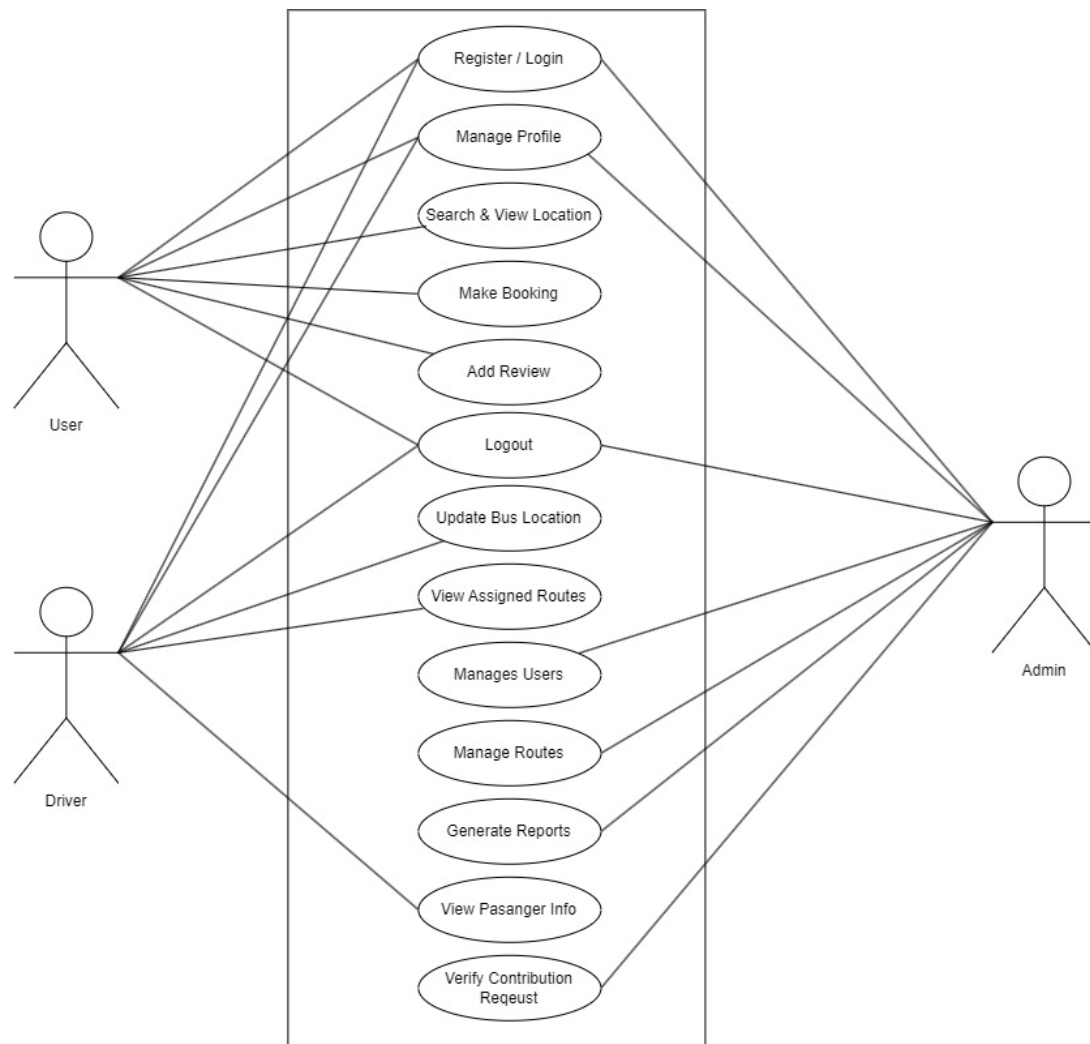


Figure 3. 1: Use Case Diagram of KHOJA

ii. Non-Functional Requirements

- Responsive design for mobile and desktop.
- Fast map rendering and route updates.
- Secure authentication and role management.
- Scalable database structure.
- Easy-to-use interface with modern UI/UX principles.

3.1.2 Feasibility Study

i. Technical Feasibility

The project is technically feasible because the required technologies are widely available and well-supported. The system is built using modern and reliable tools that are commonly used in web development. The frontend technologies include HTML, CSS, JavaScript. The backend is developed using PHP and Java, while

MySQL is used as the database management system. For mapping functionality, Leaflet.js combined with OpenStreetMap is used, and real-time updates are handled through AJAX polling and WebSockets. Additionally, modern web browsers already support essential features like geolocation APIs, interactive maps, and responsive design, making the overall implementation practical within the project scope .

ii. Operational Feasibility

The system is operationally feasible because it is designed to be simple and user-friendly for all types of users. Users interact with the system through an intuitive map-based interface, which makes navigation easy and familiar. Drivers can manage and update routes visually without requiring advanced technical knowledge. Administrators are provided with a dashboard that supports moderation, monitoring, and basic analytics. Since the interface follows patterns similar to popular navigation applications, minimal training is required for users to effectively operate the system.

iii. Economic Feasibility

The project is economically feasible because it relies mainly on open-source which significantly reduces development and operational costs. Tools such as OpenStreetMap and Leaflet.js are free to use, eliminating licensing expenses. Similarly, open-source backend frameworks and database systems further reduce cost requirements. In addition, the system can be deployed on low-cost cloud services or shared hosting platforms during its initial phase. As a result, the overall implementation remains affordable and suitable for academic and prototype-level development.

iv. Schedule Feasibility

The time schedule of the different Stages of the project can be seen below in the following Gantt Chart.

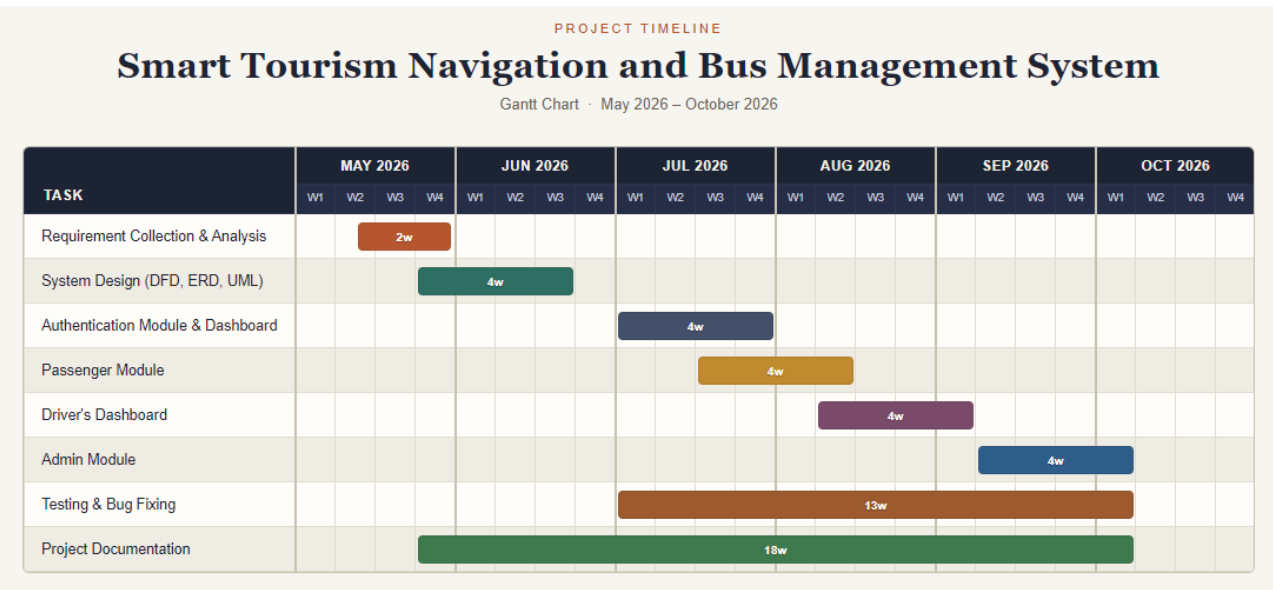


Figure 3. 2: Gantt Chart of KHOJA

3.1.3 Data Modeling

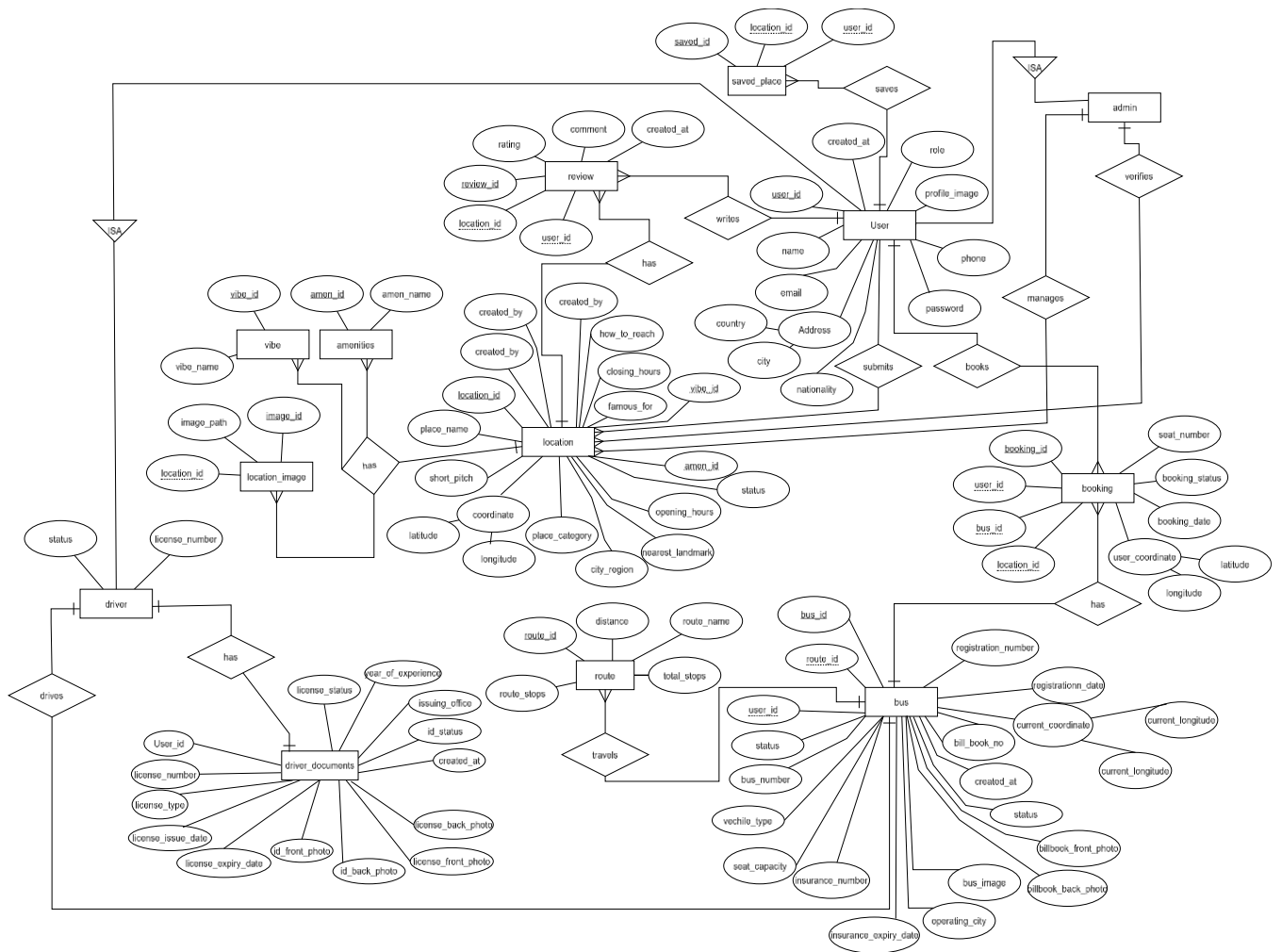


Figure 3. 3: ER Diagram of KHOJA

3.1.4 Process Modeling

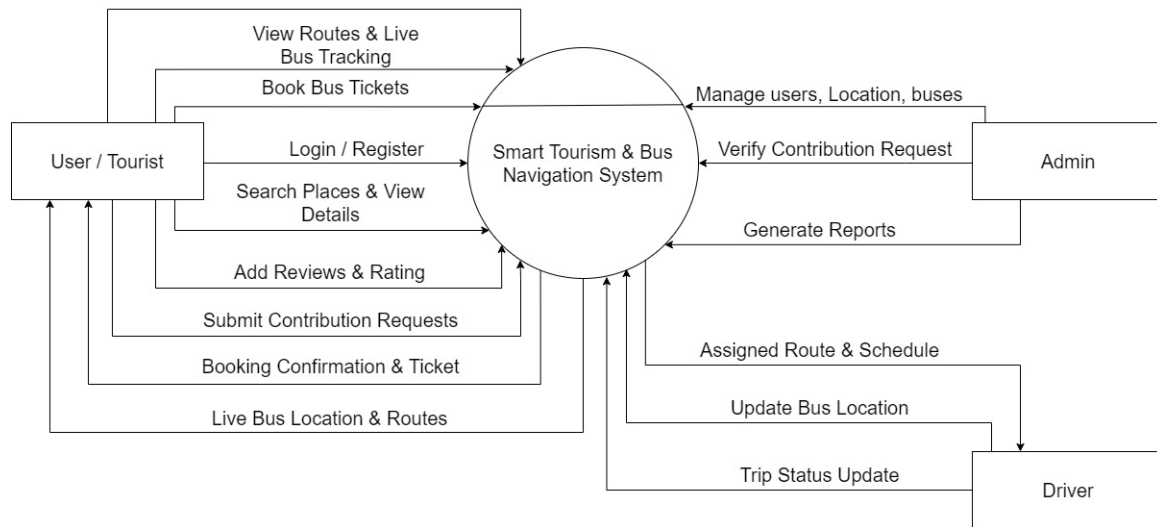


Figure 3. 4: DFD Level 0 of KHOJA

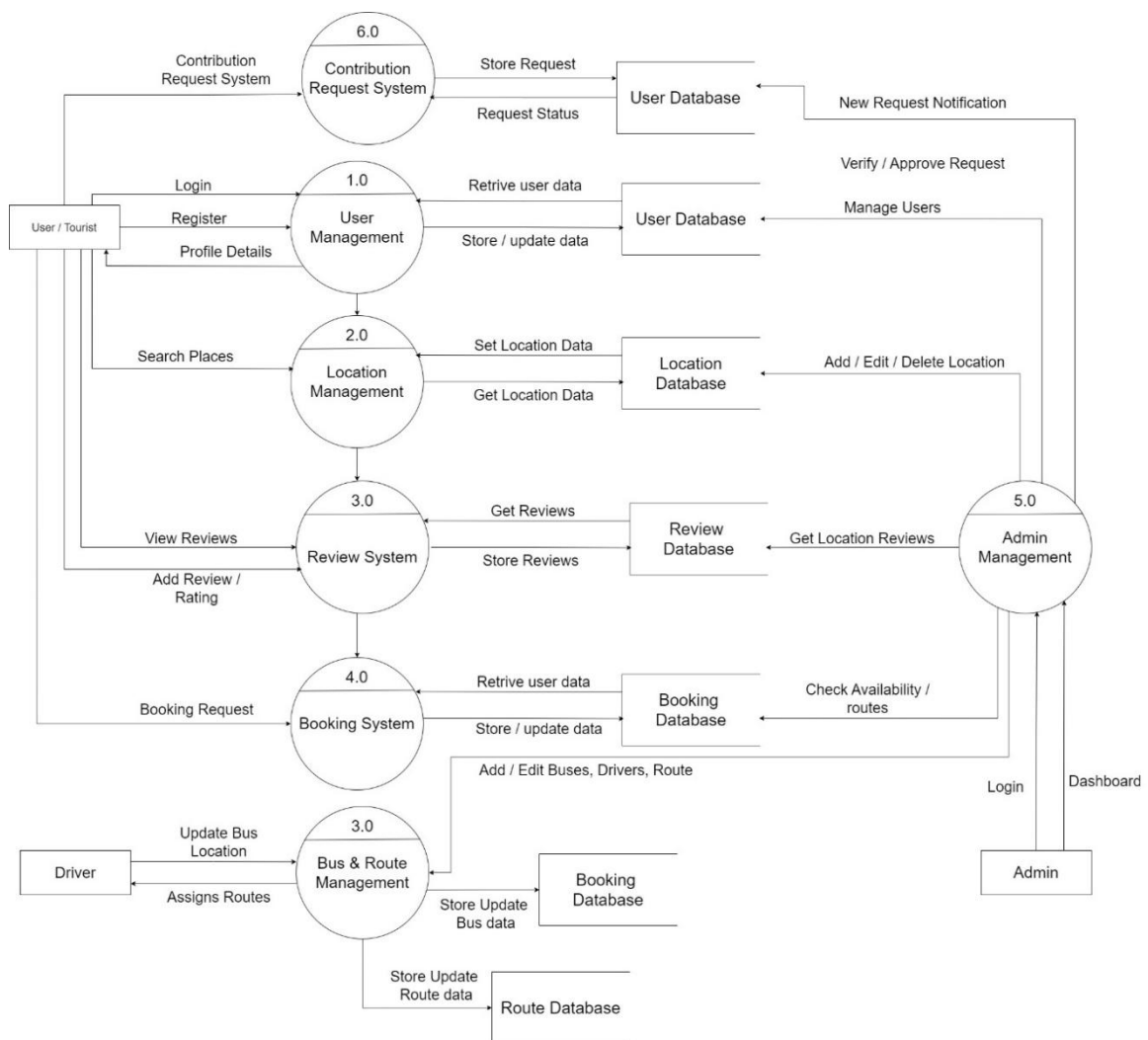


Figure 3. 5: DFD Level 1 of KHOJA

3.2 System Design

3.2.1 Architectural Design

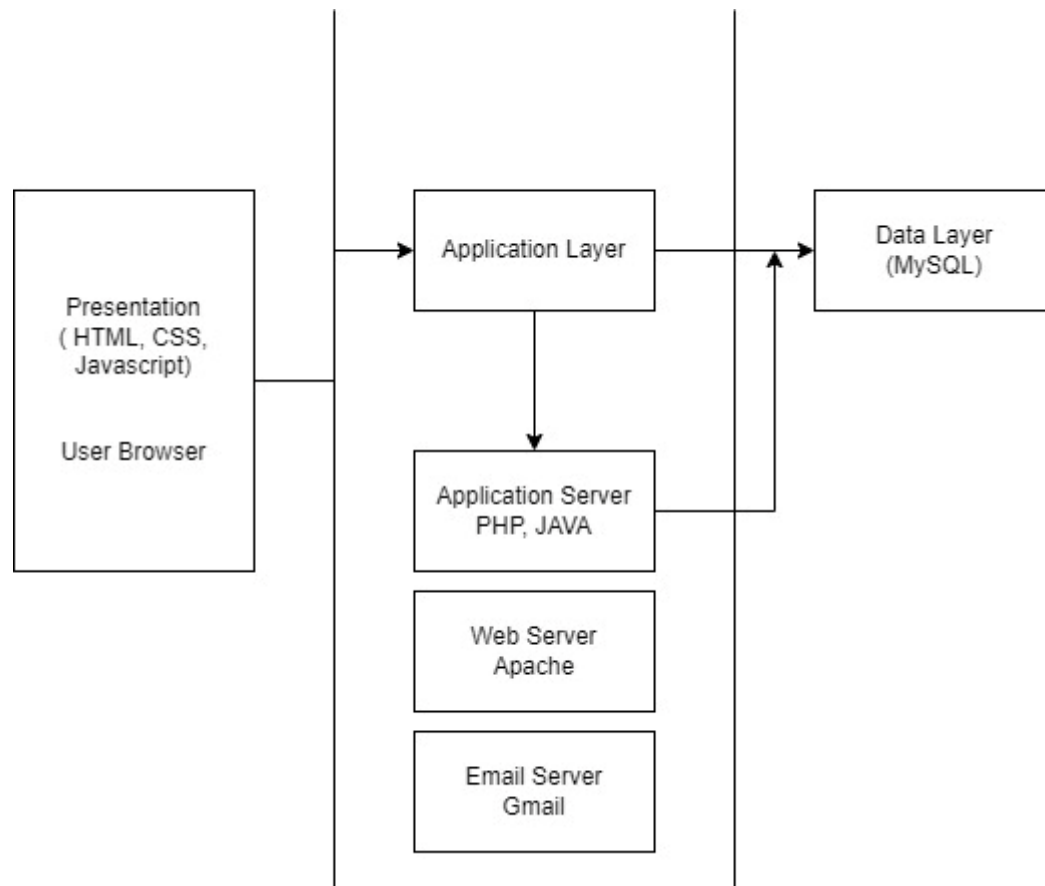


Figure 3. 6: System Architecture of KHOJA

3.2.2 Database Schema Design

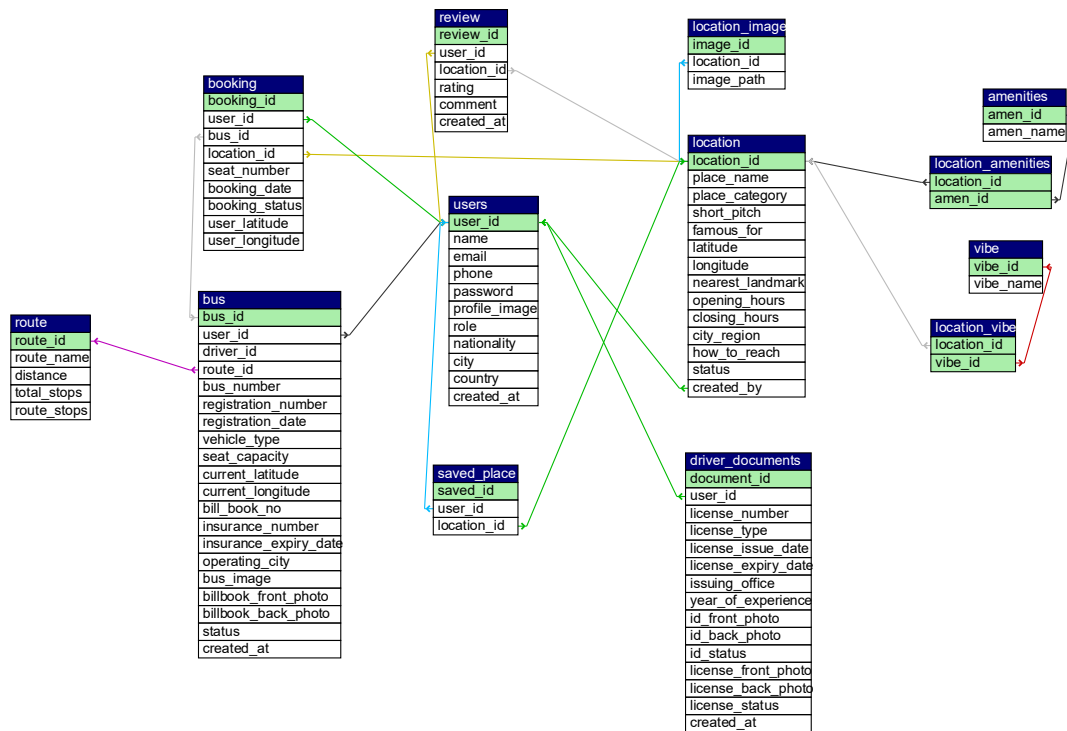


Figure 3. 7: Database Schema of KHOJA

3.2.3 Interface Design (UI / UX)

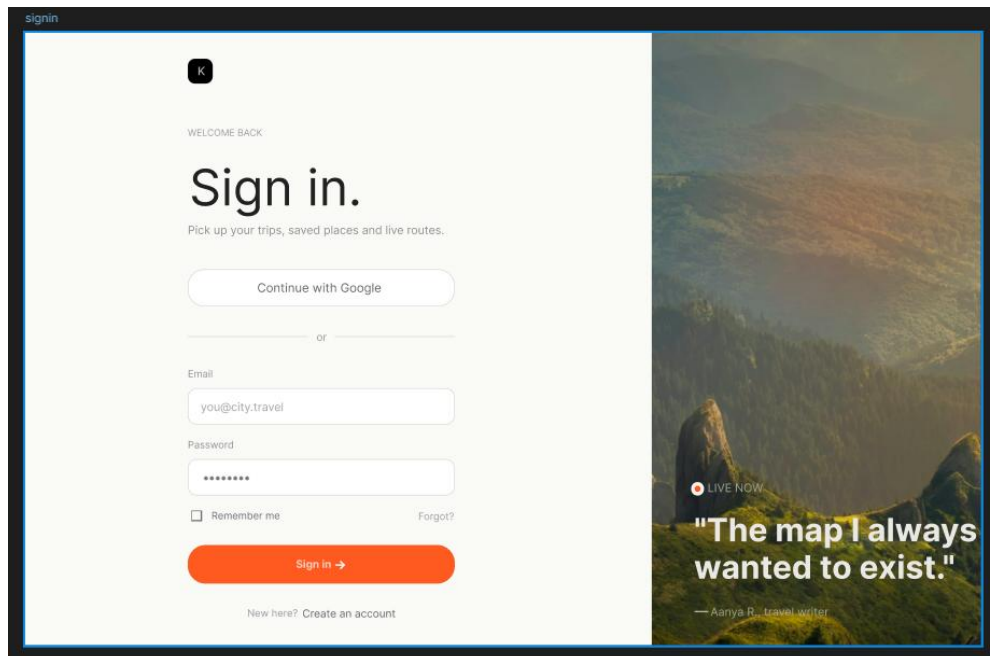


Figure 3. 8: Login Page of KHOJA

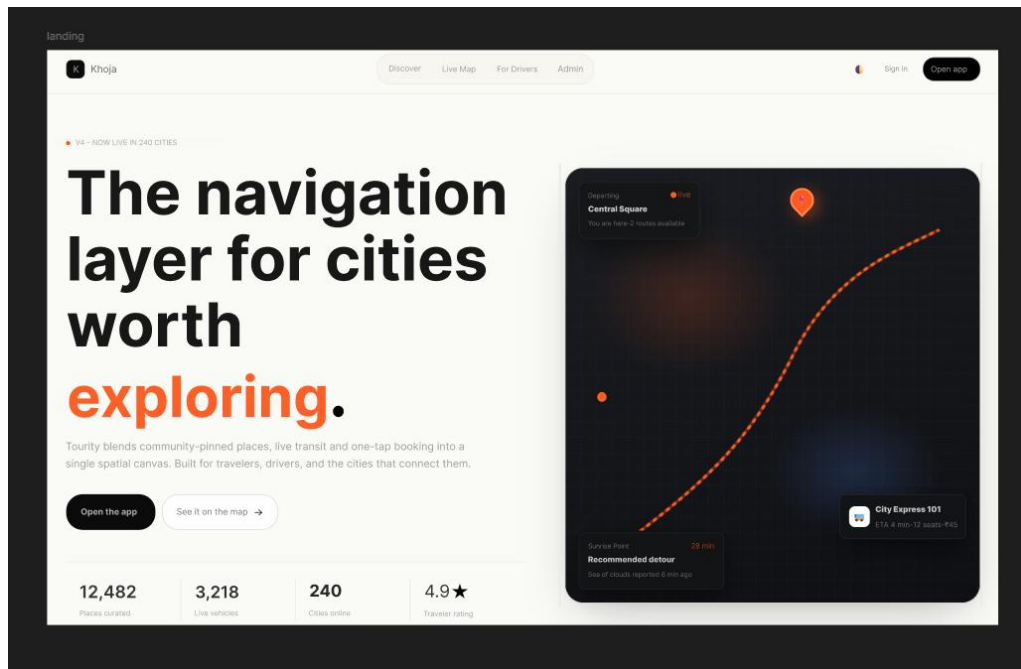


Figure 3. 9: Landing Page of KHOJA

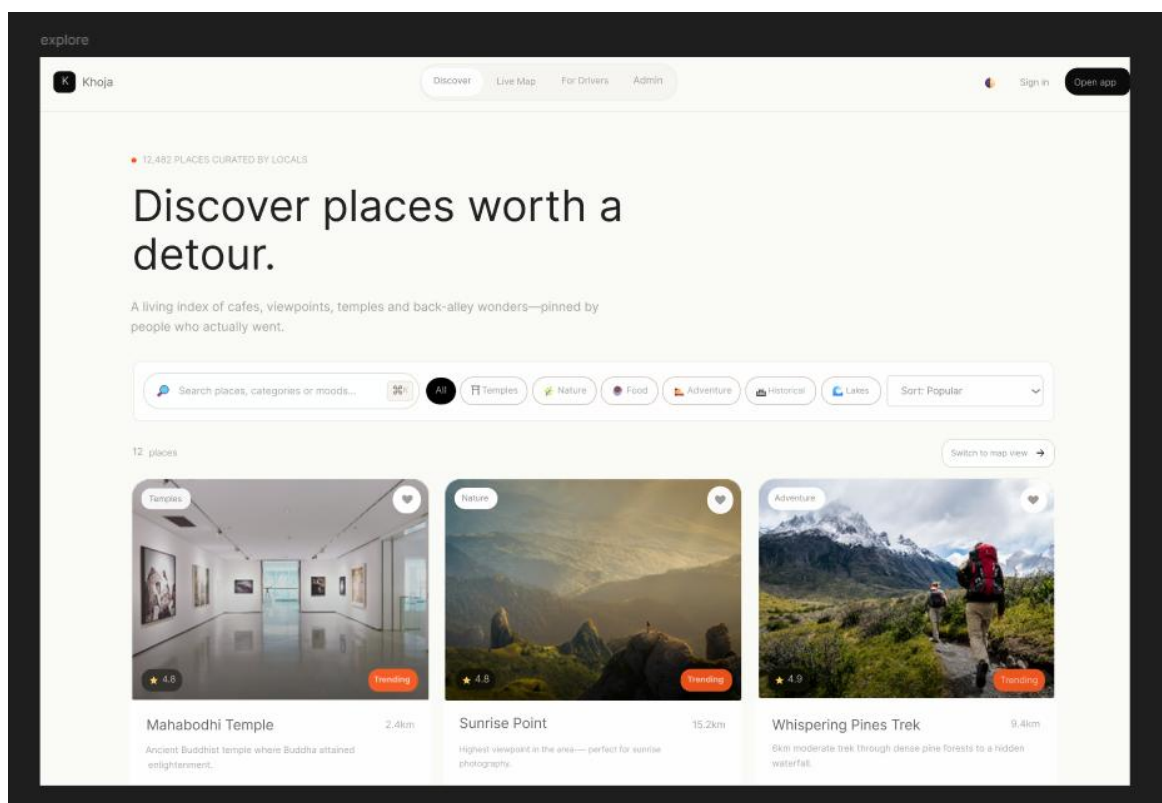


Figure 3. 10: Discover Page of KHOJA

CHAPTER 4: IMPLEMENTATION AND TESTING

4.1 Implementation

The implementation of this project focuses on developing a Smart Tourism and Navigation System that helps users discover tourist destinations, find nearby places, and navigate efficiently using public transportation. The system begins by collecting and organizing important data such as tourist locations, geographical coordinates, bus routes, bus stops, and other relevant tourism information within the targeted areas. The collected data is stored and managed in a centralized database to support efficient searching, mapping, and navigation. Using GPS and interactive map services, the system identifies nearby tourist destinations based on the user's current location and provides suitable routes to reach the selected destination.

The system also incorporates real-time bus tracking, allowing users to view available buses, their current locations, and associated routes. Route planning considers available transportation options and can suggest journeys involving bus transfers to help users reach their destination efficiently. However, the current implementation has limitations in handling complex journeys involving more than two bus transfers, as calculating and managing multiple bus switching combinations is computationally complex with the current technology stack. The platform is developed as a user-friendly web application that provides interactive maps, tourist information, bus navigation, and community-based contributions. The system aims to make tourism and public transportation more accessible and convenient by helping users explore destinations, plan their journeys, and navigate efficiently within the available transportation network.

4.1.1 Tools Used

The various system tools and technologies that have been used in developing the project, Smart Tourism and Bus Navigation System for both the front-end and backend of the project are:

FrontEnd:

- **HTML5 (HyperText Markup Language):** Used to create the structure of the website, including navigation bars, forms, cards, buttons, dashboards, maps, bus information, tourist location details, and other webpage components.

- **CSS (Cascading Style Sheets):** Used to design and style the website interface, including layouts, colors, spacing, typography, buttons, cards, responsive designs, and animations.
- **JavaScript:** Used to implement interactive functionality such as map interaction, location detection, bus and location data loading, form handling, API communication, nearby location discovery, distance calculation, and dynamic webpage updates.

Backend:

- **PHP:** Used to develop the server-side functionality of the system, including user authentication, location management, bus and booking operations, contribution handling, driver and admin operations, and communication with the MySQL database.
- **Java:** Used to develop the WebSocket server for real-time communication between drivers and users. It receives the driver's current location and transmits the location data to connected users for live bus tracking.

Database:

- **MySQL:** Used to store and manage the system's data, including user accounts, driver information, tourist locations, buses, routes, bookings, contributions, and other related records.

Version Control:

- **Git:** Used to manage and track changes made to the project's source code, maintain different versions of the system, and provide a way to restore previous versions when required.

Design & Diagram:

- **Figma:** Used during the design phase to create the user interface and user experience designs for different pages and components of the Smart Tourism and Navigation System.
- **Draw.io:** Used to create system diagrams such as the Use Case Diagram, Data Flow Diagram (DFD), flowcharts, and other system-related diagrams used in the project documentation.

4.1.2 Implementation Details of Module

- **Login Module:** The Login Module manages user authentication and access control. It allows users, drivers, and administrators to securely log in and access features according to their respective roles.
- **Location Module:** The Location Module manages tourism and transportation location data. It displays information about tourist locations, bus stops, and other relevant locations, including their geographical coordinates and related details.
- **Bus Module:** The Bus Module manages bus-related services. Users can view available buses, routes, and current bus information, as well as book buses for their journeys.
- **Map Module:** The Map Module integrates OpenStreetMap with Leaflet to display the user's current location, buses, routes, and tourist destinations. Users can interact with the map and view navigation information.
- **Contribute Module:** The Contribute Module allows users to contribute tourism-related information by submitting new locations, images, and other relevant details. Submitted contributions are reviewed before being published.
- **Discover Module:** The Discover Module helps users find nearby tourist destinations based on their current location. It uses the Haversine formula to calculate distances and recommends nearby locations accordingly.
- **Driver Module:** The Driver Module provides features for registered drivers to log in, submit required documents, manage bookings, approve passenger bookings, and operate assigned routes. Drivers can also transmit their current location through socket communication for real-time tracking.
- **Admin Module:** The Admin Module provides centralized system management. Administrators can verify driver documents and user contributions, manage users, locations, drivers, buses, bookings, and routes, and monitor the overall operation of the system.

4.2 Testing

4.2.1 Test Cases for Unit Testing

Table 4.1: Test Case of unit testing for Login Module

SN	Test Case	Test Data	Expected Result	Status
1	User login with valid credentials	Valid email and password	User successfully logs in	Pass
2	Login with incorrect password	Valid email, wrong password	Error message is displayed	Pass
3	Login with empty fields	Email and password empty	Required-field validation is displayed	Pass
4	Driver login with valid credentials	Valid driver credentials	Driver dashboard is displayed	Pass
5	Admin login with valid credentials	Valid admin credentials	Admin dashboard is displayed	Pass

Table 4.2: Test Case of unit testing for Map Module

SN	Test Case	Test Data	Expected Result	Result
1	Load interactive map	Valid map connection	OpenStreetMap map loads successfully	Pass
2	Display user's location	Valid GPS coordinates	User location is displayed	Pass
3	Zoom and pan map	User interaction	Map responds correctly	Pass
4	Load map without internet	No network connection	Map should provide an appropriate error/fallback	Pass

Table 4.3: Test Case of unit testing for Location Module

SN	Test Case	Test Data	Expected Result	Result
1	Display tourist locations	Valid location records	Locations are displayed correctly	Pass
2	Display invalid location coordinates	Invalid latitude/longitude	System should reject invalid coordinates	Fail

Table 4.4: Test Case of unit testing for Contribute Module

SN	Test Case	Test Data	Expected Result	Result
1	Submit new location	Valid location details	Contribution is submitted successfully	Pass
2	Submit location with image	Valid details and image	Location and image are submitted	Pass
3	Submit incomplete contribution	Missing required fields	Validation message is displayed	Pass
4	Submit unsupported image	Invalid file type	File should be rejected	Pass
5	Submit duplicate location	Existing location details	Duplicate contribution should be prevented	Fail

Table 4.5: Test Case of unit testing for Admin Module

SN	Test Case	Test Data	Expected Result	Result
1	Manage contributions	Pending contribution	Admin can approve or reject contribution	Pass

4.2.2 Test Cases for System Testing

Table 4.6: Test Case of System testing

SN	Test Case	Test Data	Expected Result	Result
1	Complete user registration and login	New user account	User registers and successfully logs in	Pass
2	Discover and navigate to tourist location	User location + destination	Nearby destination is displayed and navigation works	Pass
3	User contributes tourist location	Valid location contribution	Contribution reaches admin for verification	Pass
4	Admin approves contribution	Pending contribution	Approved location becomes available to users	Pass
5	Access system without internet	Network disconnected	System should handle unavailable services appropriately	Pass
6	Unauthorized user accesses admin page	Normal user account	Access should be denied and user redirected	Pass

CHAPTER 5: COMPLETED AND REMAINING TASK

5.1 Completed

- **User Authentication:** Allows users to register and log in securely to access the system's features.
- **Driver Authentication:** Allows registered drivers to log in and access driver-specific functions.
- **Admin Authentication:** Provides secure access for administrators to manage the system.
- **Interactive Map:** Integrates OpenStreetMap with Leaflet to display locations and provide an interactive map interface.
- **User Location:** Detects and displays the user's current location on the map using GPS.
- **Tourist Location Display:** Displays available tourist destinations with their relevant information.
- **Location Contribution:** Allows users to submit new tourist locations to the system.
- **Image Contribution:** Allows users to upload images along with tourist location submissions.
- **Contribution Validation:** Validates required fields and uploaded files before accepting contributions.
- **Admin Contribution Management:** Allows administrators to review, approve, and reject user-submitted locations.
- **Tourist Location Discovery:** Helps users find and navigate to tourist locations based on their current location.
- **User Access Control:** Restricts unauthorized users from accessing administrative features.

5.2 Remaining task

- **Real-Time Bus Tracking:** Allows users to view the current location of buses and receive continuous location updates.
- **Bus Booking:** Allows users to select available buses and book seats for their journey.
- **Driver Document Verification:** Allows drivers to submit required documents and administrators to approve or reject them.

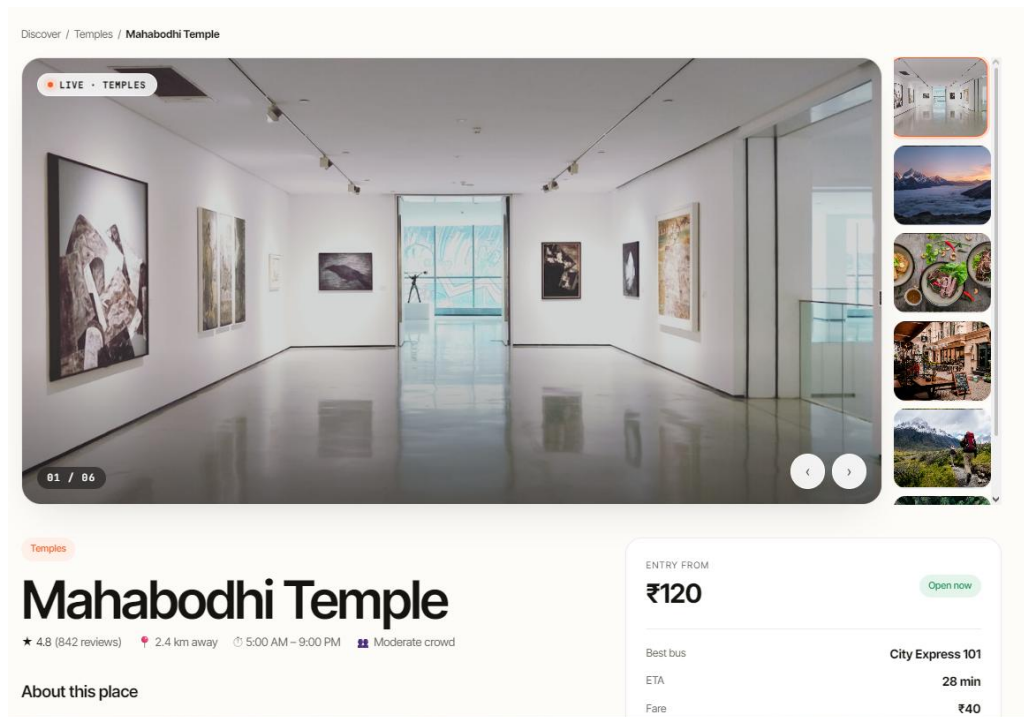
- **Driver Booking Management:** Allows drivers to view and approve passenger bookings.
- **WebSocket Location Communication:** Sends the driver's current location through socket communication and updates it for users in real time.
- **Bus Route Management:** Allows administrators to create, update, and manage bus routes and driver assignments.
- **Multi-Bus Route Navigation:** Calculates journeys requiring multiple buses and determines suitable switching points.
- **More Than Two Bus Switching:** Supports complex journeys requiring more than two bus transfers. This is currently difficult to implement efficiently with the existing technology stack.
- **Duplicate Location Detection:** Prevents users from submitting tourist locations that already exist in the system.

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APPENDICES

View Location Page



Contribute Page

K Khoja

HomeDiscoverLive MapContributeBooking

Welcome!! avashkhadkaLogout

STEP 1 of 4 EARN 50 PTS ON APPROVAL

Pin a place. Shape the map.

Four short steps. Reviewed in ~24h.

1 Basics

Name, type, vibe

2 Location

Pin on the map

3 Photos

Up to 6 images

4 Details

Hours, fees, tips

Tell us about the place

Start with the essentials. You can edit anything later.

Place Name

e.g. Hidden Waterfall on Kathmandu

Category

Nature

City / region

Kathmandu, Lalitpur

Short Pitch

Two lines that would make a friend go.

Vibe (Pick up to 4)

RomanticFamilySoloSunriseSunsetQuiteCrowdedPhoto spotFreeOffbeatLocalFavInstagrammable

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